

A Domain-Informed Hierarchical Federated Learning Framework for DDoS Detection in WSN for Critical Infrastructure

Md Facklasur Rahaman[✉], Makhduma F. Saiyed[✉], *Member, IEEE*, Irfan Al-Anbagi[✉], *Senior Member, IEEE*, and Ramakrishna Gokaraju[✉], *Senior Member, IEEE*

Abstract—The deployment of Wireless Sensor Networks (WSN) in critical infrastructure, such as Small Modular Reactors (SMRs), faces cybersecurity threats like Distributed Denial of Service (DDoS) attacks that can overload these networks and disrupt monitoring and control functions. Current DDoS detection systems often suffer from high false positive rates, neglect domain-specific operational constraints, and rely on centralized architectures that pose privacy risks, making them less suitable for distributed Internet of Things (IoT) environments. To address these issues, we propose a novel Domain-informed Hierarchical Federated Learning (DHFL) framework for WSN used in SMR monitoring and control applications. Our framework features a dual-branch bidirectional Long Short-Term Memory (LSTM) architecture comprising of two parallel processing branches with network-specific constraints, facilitating precise detection of DDoS attacks. It includes differentiable penalty functions to enforce domain-aligned behaviour and employs adaptive trust scoring to evaluate the reliability of individual nodes. These elements operate within a hierarchical Federated Learning (FL) structure organized into three tiers: sensor nodes, local aggregators, and a global coordinator, allowing collaborative training that preserves privacy. Unlike earlier approaches, our method not only maintains privacy by ensuring that raw sensor data never leaves the local nodes and only model updates are shared but also considers the operational importance and trustworthiness of each node through tier-weighted aggregation. Tested on the CICIOT2023 dataset, our system achieved 93.4% accuracy, 94.5% precision, 97.5% recall, 95.5% F1-score, and 98.9% AUC, surpassing state-of-the-art FL methods in both performance and efficiency. Furthermore, it converged in fewer communication rounds (30–50) with reduced communication costs (from 45 MB to 30 MB per round). Our framework can differentiate between normal reactor transients and actual attacks, making it suitable for mission-critical SMR cybersecurity.

Index Terms—Wireless sensor networks (WSN), small modular reactor (SMR), distributed IoT sensors, federated learning, LSTM, hierarchical aggregation, DDoS attack detection, domain-informed LSTM, trust-aware systems.

I. INTRODUCTION

WIRELESS Sensor Networks (WSN) have emerged as fundamental building blocks for modern Internet of Things (IoT) systems. They enable real time monitoring and control across diverse industrial applications through distributed sensing capabilities and autonomous data collection mechanisms. These networks, consisting of spatially distributed sensor nodes that communicate wirelessly to monitor physical or environmental conditions, have proven valuable in mission-critical infrastructures where continuous monitoring and rapid response capabilities are essential for operational safety [1]. These networks have found widespread adoption across various sectors including energy industry automation, transportation systems, and critical infrastructure protection, demonstrating their versatility and reliability in mission-critical applications [2].

The use of WSNs in critical infrastructure such as Small Modular Reactors (SMR)-based power generation units facilitate real-time condition monitoring, support safety, and enable rapid decision-making [3]. As critical infrastructure components, SMR-based power generation units rely on WSN to enable real-time monitoring of reactor parameters, coolant systems, and safety mechanisms across distributed facility locations. However, these technological advances also expose SMRs to a variety of cyber threats, particularly in terms of potential disruptions that could compromise the integrity and safety of these critical systems.

Among the various cyber threats facing SMR-based power generation units, Distributed Denial of Service (DDoS) attacks pose a significant risk due to their potential to overwhelm the WSN enabled systems [4]. In such environments, WSN deployments create distributed attack surfaces where DDoS threats can target individual sensor nodes, communication gateways, or the entire network. Typically initiated by botnets or malicious entities, these attacks flood the network with traffic, disrupting real-time monitoring, control, and safety operations by congesting sensor communication links and data aggregation points [5]. The distributed nature of WSN makes

Received 24 December 2025; revised 8 April 2026; accepted 8 May 2026. Date of publication 13 May 2026; date of current version 19 May 2026. This work was funded by the Natural Sciences and Engineering Research Council of Canada and Canadian Nuclear Safety Commission as the Small Modular Reactors Research Grant, ALLRP 580480-2022. The associate editor coordinating the review of this article and approving it for publication was K. Hammar. (Corresponding author: Irfan Al-Anbagi.)

Md Facklasur Rahaman and Irfan Al-Anbagi are with the Faculty of Engineering and Applied Science, University of Regina, Regina, SK S4S 0A2, Canada (e-mail: facklasurrahaman@gmail.com; Irfan.Al-Anbagi@uregina.ca).

Makhduma F. Saiyed is with the Department of Computer Science, Trent University, Oshawa, ON L1J 5Y1, Canada (e-mail: makhdumabanusaiyed@trentu.ca).

Ramakrishna Gokaraju is with the Department of Electrical and Computer Engineering, University of Saskatchewan, Saskatoon, SK S7N 5A9, Canada (e-mail: rama.krishna@usask.ca).

Digital Object Identifier 10.1109/TNSM.2026.3693112

them particularly vulnerable, as attackers can simultaneously target multiple sensor clusters to amplify disruption effects.

For instance, one of the most notable DDoS attacks takes place in 2022, when Ukraine's state-owned nuclear power company, Energoatom, reports an unprecedented DDoS attack orchestrated by the hacktivist group People's Cyber Army. The group employs a botnet of 7.25 million compromised accounts to flood Energoatom's website, rendering it temporarily inaccessible. Despite the scale of the attack, Energoatom quickly regains control and no operational disturbances are reported at the nuclear facilities [6]. Additionally, the United Kingdom's Sellafield nuclear waste site experiences cyberattacks. While these attacks did not directly damage critical systems, they underscore the persistent threat to critical infrastructure facilities and their supporting WSN architectures [7]. In 2021, a cybercriminal group, Garnesia Team, claimed responsibility for a DDoS attack on the Nuclear Power Corporation of India Limited (NPCIL) website. The attack results in temporary inaccessibility of the site, although no impact on operational systems is confirmed [8]. Before North Korea's nuclear test in 2016, a DDoS attack targeted a website that monitors the country's nuclear test site. The timing of the attack raises suspicions about its origin, although there is no direct evidence linking it to state-sponsored actors [9].

To detect various cyberattacks, several detection approaches have been developed using Machine Learning (ML) and Deep Learning (DL) models. These models are trained on historical data to recognize and classify malicious traffic. However, existing solutions face multiple limitations for critical infrastructure [10]. Traditional Intrusion Detection Systems (IDS) often produce high false positives in dynamic environments [11], which is unacceptable for critical infrastructure where false alarms can lead to costly shutdowns and missed detections can pose radiological risks [12]. These systems also lack integration of the domain knowledge of the operational network, leading to misclassification of legitimate transitions of the state of the reactor as anomalies and the failure to detect attacks that breach the specific time and safety constraints of the SMR [13]. Furthermore, the heterogeneity of sensor communication patterns in WSN-based architectures, which varies with physical location and operational criticality, makes detection more challenging.

To overcome the limitations of conventional ML and DL models in specialized critical infrastructure, researchers have explored domain-informed approaches that embed specific operational constraints into the detection process. One such model, the Domain Informed Long Short-Term Memory (LSTM), integrates physics-based rules such as permissible packet rates, bandwidth thresholds, and thermal safety limits into its architecture [14]. This design helps reduce false positives by aligning model behavior with actual operation conditions, allowing it to better differentiate between normal system state transitions and malicious traffic. However, while domain-informed models improve context awareness and detection accuracy, they still rely on centralized architectures and require access to large volumes of sensitive operational data, which introduces privacy risks and communication overhead.

To address the privacy and communication limitations of centralized domain-informed models, Federated Learning (FL) has emerged as a promising approach that enables decentralized model training directly on edge devices [15]. This eliminates the need to transmit sensitive raw data, reducing communication overhead and privacy risks [16], [17]. FL allows distributed devices to collaboratively train models while sharing only model updates with a central aggregator [18]. However, conventional FL frameworks also exhibit critical limitations in such networks. Specifically, they fail to address the heterogeneous and multi-tiered architectures, where certain tiers require a higher level of trust and influence compared to peripheral monitoring devices. Moreover, most existing FL methods do not incorporate operational domain constraints, which are essential to reliably distinguish between normal behavior and malicious activity.

To address these challenges, this paper proposes a Domain-informed Hierarchical Federated Learning (DHFL) framework, designed for cyberattack detection in critical infrastructure. The DHFL framework enables collaborative model training across distributed sensor nodes without sharing raw data, thus preserving privacy and reducing communication overhead. The framework integrates a domain-informed BiLSTM model with a hierarchical, trust-aware aggregation mechanism that reflects the varying criticality of the nodes within the networks. The core novelty lies in its dual-branch LSTM architecture (DI-LSTM), which concurrently captures temporal network traffic patterns and specific operational constraints through a differentiable penalty function. This design enables DHFL framework to accurately distinguish between legitimate operational transients from reactors and actual cyberattacks. By embedding domain knowledge directly into both the model architecture and the training process, the proposed system overcomes the limitations of traditional IDS approaches, which often generate high false positives by misclassifying routine state changes as anomalies. The primary contributions of this paper are as follows:

- We propose a domain-informed dual-branch Bi-LSTM model for detecting DDoS attacks in IoT networks. This model integrates temporal traffic pattern learning with domain constraints such as packet rate limits and control system latencies. Embedding these constraints as differentiable penalty functions improves detection accuracy and reduces false positives by enforcing physically consistent behavior.
- We develop a hierarchical FL framework (DHFL) to preserve data privacy and address node heterogeneity that groups nodes by criticality into core, edge, and peripheral tiers based on their influences. An adaptive trust scoring mechanism evaluates nodes across multiple dimensions to guide participant selection and trust-weighted aggregation, thus improving robustness against malicious clients.
- We incorporate secure aggregation protocols to protect local model updates during federated training. This ensures a resilient and privacy-preserving detection framework, tailored for the stringent cybersecurity requirements of mission-critical SMR environments.

The remainder of this paper is structured as follows: Section II provides an overview of the related work. Section III presents a detailed explanation of the proposed methodology. Section IV elaborates on the performance evaluation of the overall approach. Finally, Section V concludes the paper and outlines the directions for future work.

II. RELATED WORKS

This section reviews the state-of-the-art in detecting DDoS attacks, with particular emphasis on ML, DL and FL frameworks in cyber-physical systems. Recent advances in FL techniques aimed at preserving data privacy and addressing heterogeneous, resource-constrained environments are critically examined. Additionally, the emergence of physics-informed DL models which integrate domain-specific operational constraints to enhance anomaly detection accuracy and interpretability, is discussed.

The development of DDoS detection has gained significant attention in recent years due to its criticality in ensuring the security and privacy of critical and sensitive infrastructures. The authors in [24] proposed a holistic feature engineering and ML framework for detecting DDoS attacks, which simulated cross-layer DDoS attacks in standardized IoT networks based on the 6LoWPAN protocol stack. It achieved enhanced detection accuracy and reduced computational overhead through optimized feature selection and ensemble classification. The proposed approach relied on offline training and centralized detection on the 6LoWPAN Border Router, which could become a single point of failure and faced practical challenges in real-world deployment related to sniffer node placement and communication reliability. In [22], the authors proposed a time-efficient Software-Defined Networking (SDN)-based security framework using a customized SDNWISE controller to detect DDoS attacks in IoT networks through session IP counters and IP payload analysis. The framework achieved early-stage detection with high accuracy (98%–100%) and low false positive rates. However, it relied on a centralized controller and logging mechanism, which could become a single point of failure and face scalability issues in large, diverse IoT deployments. Similarly, in [21], the authors introduced an explainable hybrid DL model that combined convolutional neural network (CNN) and LSTM for real-time DDoS detection in SDN-IoT-based Healthcare Industry 5.0 environments. They used SHapley Additive exPlanations (SHAP) for interpretability and showed high detection accuracy on benchmark datasets. However, the model's dependence on supervised learning and large amounts of training data might limit its ability to adapt to new or evolving attack types in dynamic healthcare IoT systems.

Building upon these advancements, recent research has explored the integration of domain-specific knowledge into ML models to enhance detection robustness and interpretability in critical IoT environments and infrastructures. Among these, the Physics-Informed DL model stands out by embedding physical constraints directly into the detection process, addressing key challenges in safeguarding sensitive and safety-critical networks. Likewise, Wu et al. [14] developed a

physics-informed recurrent neural network (PIRNN) detector combined with a knowledge-guided extended Kalman filter (EKF) for real-time detection and resilient control of sensor cyberattacks in nonlinear chemical processes. This approach showed improved detection accuracy, lower data requirements, and better state estimation compared to traditional data-driven methods. Additionally, Wu et al. [23] proposed a physics-informed graph convolutional recurrent network (PIGCRN) that incorporated chemical process topology and prior knowledge of cyberattack patterns into a unified spatial-temporal learning framework. This model improved detection accuracy in complex chemical process networks while reducing the need for large training datasets. However, both methods depended on accurate prior knowledge of attack patterns and system-specific domain information, which could limit their effectiveness against unknown or evolving cyberattacks and in cases with incomplete or inaccurate process models.

Moreover, Vyas et al. [25] developed a physics-informed neural network (PINN) framework for cyberattack detection in vehicle platoons which combines physics-based models and real-time data to improve attack isolability and reduce the effect of model uncertainties. Gaggero et al. [26] provided a survey of AI-based and physics-based anomaly detection methods in the Smart Grid, highlighting the integration of physical laws with ML to improve detection robustness and resilience. Lei et al. [27] proposed a physics-informed LSTM with adaptive weight allocation for accurate and stable prediction of dissolved gases in power transformers. Their method combined data-driven and physical models to enhance forecast accuracy under normal and fault conditions. However, a common limitation in these works was the reliance on centralized data collection, raising privacy concerns and limiting scalability and collaborative learning in real-time between distributed, privacy-sensitive nodes.

Recent advances in FL and physics-informed ML have demonstrated promising directions for secure, scalable, and privacy-preserving anomaly detection in heterogeneous IoT environments. Iqbal et al. [20] proposed a hierarchical continual FL framework for healthcare Internet of Medical Things (IoMT) applications, integrating continual learning strategies with FL to overcome catastrophic forgetting while ensuring privacy and domain knowledge retention through multi-tier models. Wu et al. [28] addressed the challenge of non-Independent and Identically Distributed (non-IID) data distribution in IoT by developing an improved Gale–Shapley-based hierarchical FL method, which pairs devices with complementary datasets to improve local model quality, thus improving global convergence speed and accuracy without additional communication overhead. Similarly, apart from cyberattack detection, in other target systems, Devaraj et al. [29] presented a hierarchical FL architecture designed for precision agriculture, combining distributed sensing with a personalized FL strategy to enable scalable, privacy-aware crop health monitoring across geographically dispersed farms. Wang et al. [19] introduced a two-tier hierarchical FL scheme supported by aerial relay nodes that serve as both data sources and intermediate aggregators, optimizing resource allocation and minimizing latency for FL convergence in wireless IoT

TABLE I
COMPARATIVE ANALYSIS OF PROPOSED DHFL FRAMEWORK WITH EXISTING WORKS

Existing Works	Target System	Employed Technique	Dataset	Hierarchical FL Structure	Adaptive Client Trust Scoring	Handling Non-IID and Node Heterogeneity	Multi Tier Node Organization	Domain Informed Local Model
[19]	Wireless IoT networks with UAV relays	FL	Simulated wireless IoT network datasets	✓	×	Partial	×	×
[20]	IoMT healthcare applications	FL+Continual Learning	Medical imaging datasets	✓	Partial	✓	✓	×
[21]	SDN-enabled Healthcare Industry 5.0 IoT	CNN+LSTM	CICDDoS2019, IoT healthcare security datasets	×	×	×	×	×
[22]	SDN-based IoT networks	ML (Counter-based, session IP analysis)	SD-IoT testbed simulation data	×	×	×	×	×
[23]	Nonlinear chemical process network	GCN+LSTM	Synthetic dataset generated via Aspen Plus Dynamics simulation	×	×	×	×	✓
DHFL	SMR sensor networks	FL	CICIoT2023 dataset	✓	✓	✓	✓	✓

networks, thus improving system robustness and communication efficiency. Despite these contributions, the presented the approaches largely depended on handling heterogeneous, dynamic, and often non-IID data under strict resource and privacy constraints. This makes real-time model adaptation and generalization difficult, especially in privacy-sensitive and safety-critical IoT applications. Moreover, the lack of explicit integration of domain-specific physical constraints in anomaly detection models often results in higher false positive rates and increased vulnerability to subtle, sophisticated attacks that closely mimicked normal behavior.

Recent research efforts have begun to address the cybersecurity of SMR-specific control systems, with a focus on both architectural and detection model innovations. In [13], Salehpour and Al-Anbagi developed a cyberattack detection system for SMR-based digital substations, integrating real-time simulation using RTDS and NS3 to evaluate the performance of machine learning-based detection. However, their framework lacked domain-informed context and did not include any privacy-preserving mechanisms. Ayodeji et al. [30] introduced a wave-attention DL model for real-time attack detection in digital SMR control systems using the Asherah Nuclear Simulator. The model demonstrated higher accuracy compared to traditional approaches; however, it relied on centralized data processing and did not incorporate federated or trust-aware learning. Rockabrand's dissertation [31] proposed the integration of AI into SMR emergency preparedness frameworks, focusing on real-time monitoring and autonomous response. However, it highlighted key challenges in aligning AI with operational safety of SMR and human decision-making. Meanwhile, Nam et al. [32] introduced a high-fidelity testbed for cyber-physical SMR systems designed to simulate and evaluate advanced attacks. Despite its capabilities, the testbed lacked a federated architecture and a hierarchical defense model. These studies reflected growing interest in SMR-specific cybersecurity but shared common limitations in privacy preservation, multi-tier node organization, and the integration of domain-informed anomaly detection. Table I shows a comparative analysis of recent

frameworks, highlighting key limitations such as the lack of hierarchical FL, adaptive trust, and domain-informed models. In contrast, our proposed framework addresses these gaps through a domain-aware hierarchical FL approach for SMR sensor networks.

Our proposed DHFL framework directly addresses the challenges of high false positive rates when legitimate reactor state transitions (such as startup sequences or emergency procedures) are misclassified as cyberattacks and the lack hierarchical FL structures that account for the different operational importance levels in SMR systems, which fails to give critical reactor control nodes greater influence than peripheral monitoring systems while maintaining privacy across distributed sensor networks. It does that by embedding network domain system constraints into a dual-branch DI-LSTM architecture for temporal and domain-informed feature learning at the local node level. This integration enforces physically domain-consistent behavior during detection, substantially reducing false positives and enhancing generalization to unseen attack patterns. Furthermore, the hierarchical FL structure orchestrates collaborative model training by partitioning SMR IoT sensor networks into core, edge, and peripheral tiers, with an adaptive trust scoring mechanism evaluating nodes across multiple dimensions to ensure robust aggregation that is resilient to malicious or low-quality updates. This tiered approach balances computational load and communication overhead, enabling scalable, privacy-preserving, and upgraded DDoS attack detection within resource-constrained mission-critical environments. By coupling domain-informed local learning with trust-aware hierarchical FL, our DHFL framework overcomes the inherent limitations of existing IoT anomaly detection frameworks, providing a logically and technically sound solution to safeguard critical infrastructure in complex, dynamic settings.

III. THE PROPOSED DHFL FRAMEWORK WITH WORKFLOW

Before presenting our DHFL framework architecture, it is essential to establish the foundational context of SMR

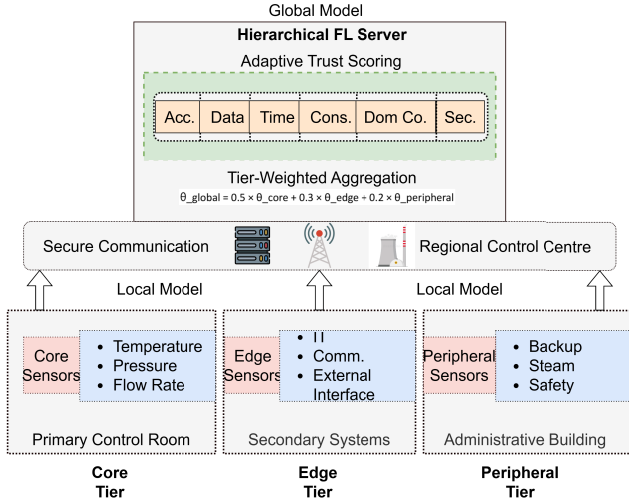


Fig. 1. DHFL framework for DDoS detection in SMR sensor networks with tier-weighted aggregation.

operational environments and their cybersecurity requirements. This preliminary discussion provides the necessary background to understand why conventional FL approaches are insufficient for critical infrastructure and how our domain-informed hierarchical design addresses the specific challenges of SMR sensor networks. The following subsections detail the operational constraints, security hierarchies, and trust dynamics that inform our DHFL framework design decisions.

A. WSN-Based SMR Monitoring

Unlike conventional IoT environments, SMR IoT networks operate under stringent safety-critical constraints where sensor data directly influence reactor control decisions, necessitating detection systems that inherently understand physical operational boundaries. We consider a hierarchical FL structure with three distinct tiers core (reactor control systems), edge (secondary monitoring), and peripheral (facility networks) which mirror the inherent security classification and operational criticality within SMR where core systems demand the highest trust and computational priority while peripheral systems provide supplementary intelligence without compromising critical operations. This tiered approach becomes essential given that SMR are typically deployed in geographically isolated, high-security zones where centralized data aggregation poses both logistical and cybersecurity risks, making FL not only advantageous, but operationally necessary. The integration of Domain-Informed LSTM (DI-LSTM) over traditional LSTM architectures addresses a fundamental limitation in cyber-physical attack detection: conventional models may flag legitimate operational transients (such as reactor startup sequences or emergency response procedures) as anomalies, whereas DI-LSTM incorporates domain-specific constraints as differentiable penalty functions within the dual-branch architecture to distinguish between benign operational variations and genuine cyberattacks. The adaptive trust scoring mechanism becomes critical in this context because SMR nodes exhibit heterogeneous reliability profiles based on their physical location, security clearance levels, and exposure to

electromagnetic interference. Figure 1 illustrates the proposed DHFL framework.

DHFL contains a dual-branch bidirectional LSTM architecture that simultaneously processes temporal network traffic patterns and SMR-specific operational constraints through differentiable penalty functions, integrated within a hierarchical FL structure with adaptive trust scoring. Initially, distributed SMR sensor nodes (organized into core, edge, and peripheral) locally train dual branch bidirectional LSTM models on private datasets. It also simultaneously computes domain compliance penalties through domain-informed constraints in the network. This can distinguish legitimate reactor operational transients from genuine cyberattacks. These nodes then transmit model parameters and trust metrics to a central aggregator which evaluates multidimensional trust scores across six dimensions (accuracy, data quality, timeliness, consistency, domain compliance, security clearance) and performs tier-aware node selection, which ensures balanced representation across operational hierarchies. For our framework, we choose these six dimensions because they can capture both computational readability and operational integrity. Computational readability tracks four dimensions, which are accuracy, data quality, timeliness, and consistency, whereas operational integrity adds domain compliance and security clearance assessments. This is crucial for SMR environments where individual nodes show different reliability patterns. The placement of the nodes varies based on their physical location, security level requirements, and exposure to electromagnetic interference. These factors can degrade specific performance metrics, making trust evaluation across those dimensions essential for maintaining the framework's integrity. The aggregation proceeds through a two-level scheme: intra-tier trust-weighted averaging followed by inter-tier combination using fixed weights ($core = 0.5$, $edge = 0.3$, $peripheral = 0.2$), preserving the strategic influence of critical reactor control systems while incorporating intelligence from secondary monitoring and facility networks. Finally, the global model is distributed back to participating nodes, enabling continuous adaptation to evolving threats while maintaining network domain-informed constraints essential for SMR operational integrity, creating a hierarchically aware, privacy-preserving defense framework.

B. Problem Formulation

The problem of DDoS attack detection in SMR sensor networks can be formally characterized as a multi-objective optimization challenge that must simultaneously address anomaly detection accuracy, privacy preservation, and operational integrity under safety-critical constraints. Let $\mathcal{N} = \{N_1, N_2, \dots, N_n\}$ represent the set of distributed SMR sensor nodes organized into three hierarchical tiers $\Theta = \{\Theta_{core}, \Theta_{edge}, \Theta_{peripheral}\}$, where each node N_i maintains a local dataset $\mathcal{D}_i = \{(x_{i,j}, y_{i,j})\}_{j=1}^{|\mathcal{D}_i|}$ containing network traffic features and binary labels (DDoS/Normal). The primary objective is to learn a global detection model θ_{global} that minimizes the federated loss function:

$$\min_{\theta_{global}} \sum_{i=1}^n \omega_i \cdot \tau_i \cdot \mathcal{L}(f_{\theta}(\mathbf{X}_i), \mathbf{Y}_i) + \lambda \cdot \Phi(\mathbf{X}_i) \quad (1)$$

subject to the constraints: Raw data $\mathcal{D}_i$ never leave node N_i , with only model parameters $\theta_i^{(t)}$ transmitted during federated rounds; All network features must satisfy SMR operational bounds; Node influence weights ω_i must reflect both tier membership ($\omega_{\text{tier}} \in \{0.5, 0.3, 0.2\}$) and adaptive trust scores $\tau_i = \sum_{d=1}^6 \omega_d \cdot s_{i,d}$ across six dimensions; and Local datasets exhibit heterogeneous class distributions $|\text{DDoS}_i|/|\text{Normal}_i| \neq \text{constant}$, reflecting real-world deployment scenarios where different SMR subsystems experience varying attack patterns.

The dual-branch domain-informed architecture addresses the fundamental limitation that conventional LSTM models cannot distinguish between legitimate operational transients (reactor state transitions, emergency procedures) and malicious anomalies, as both may exhibit similar statistical signatures but differ in their adherence to reactor constraints. The domain penalty function $\Phi(P, B, S) = (\varphi_{\text{packet}} \cdot \varphi_{\text{bandwidth}} \cdot \varphi_{\text{size}})^{1/3}$ acts as a domain-informed regularizer that penalizes violations of SMR communication constraints, thereby reducing false positive rates. The hierarchical FL framework with tier-weighted aggregation $\theta_{\text{global}}^{(t+1)} = \sum_{\text{tier}} \omega_{\text{tier}} \cdot \theta_{\text{tier}}^{(t+1)}$ ensures that critical reactor control nodes (Core Tier) maintain disproportionate influence over the global model while still incorporating intelligence from secondary monitoring systems (edge tier) and administrative facility networks (peripheral tier), thus balancing detection accuracy with operational safety priorities. This formulation transforms the traditionally centralized DDoS detection problem into a distributed, privacy-preserving solution that explicitly accounts for the heterogeneous, mission-critical nature of the SMR cyber-physical infrastructure.

C. Temporal-Aware LSTM Modeling for Traffic Dynamics

Building upon the formalized optimization framework, the detection of DDoS attacks in SMR IoT sensor networks requires the capture of rapid, temporal deviations in otherwise consistent communication flows that characterize normal reactor operations. These time-sensitive anomalies manifest as sudden changes in packet transmission patterns, spikes in bandwidth utilization, or unusual latency variations that can compromise critical control communications. To address this temporal complexity, we employ a neural network capable of learning temporal dependencies in both forward and backward directions.

The LSTM architecture is governed by gated operations that regulate the flow of information across time steps, dynamically updating the memory and hidden states of the network to capture both long-term operational baselines and short-term anomalous deviations. The functional components of the LSTM cell are defined as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \quad (2)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \quad (3)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C), \quad (4)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (5)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \quad (6)$$

$$h_t = o_t \odot \tanh(C_t), \quad (7)$$

where x_t represents the input feature vector at time t , h_t and C_t denote the hidden and cell states respectively, $W_{\{f,i,C,o\}}$ and $b_{\{f,i,C,o\}}$ are learnable weights and biases, $\sigma(\cdot)$ is the sigmoid activation function, and $\odot$ indicates element-wise multiplication.

To enrich the model's representation capacity for detecting bidirectional attack patterns, a bidirectional configuration is employed where the output at each time step t is a concatenation of forward and backward hidden states:

$$h_t = [\vec{h}_t; \overleftarrow{h}_t]. \quad (8)$$

The Bi-LSTM architecture implemented in this study comprises two layers optimized for SMR network traffic characteristics. The first layer contains 128 bidirectional units with return sequences enabled, followed by batch normalization and dropout (rate = 0.4) to prevent overfitting. The second layer consists of 64 bidirectional units without sequence return, again followed by batch normalization and dropout (rate = 0.4). These stacked layers allow the model to abstract multi-level temporal features from the input traffic flows, progressing from low-level packet-by-packet patterns to high-level communication flow signatures. Subsequent dense layers perform nonlinear transformations for binary classification, with dropout rates of 0.3 and 0.2 respectively to maintain generalization capability.

This temporal encoder serves as the foundation for the domain-informed dual-branch design, which further incorporates domain-specific physical constraints of SMR systems to distinguish between legitimate operational transients and genuine cyberattacks, as detailed in the following subsection.

D. Domain-Informed Dual-Branch LSTM Architecture

Building upon the foundation of temporal modeling established in the previous subsection, we enhance the robustness of the detection framework by extending the core Bi-LSTM network with a domain-informed module that integrates domain-specific constraints from SMR operations, as shown in Figure 2. This dual-branch architecture enables the model to process temporal traffic features and physically grounded attributes in parallel, enforcing structural priors during learning, and reducing susceptibility to adversarial or physically implausible inputs that could compromise detection accuracy in safety-critical SMR environments.

The operational characteristics of SMRs impose network domain constraints on communication behavior, particularly in the dynamics of data transmission rates, packet sizes, and bandwidth usage [33]. These constraints are derived from safety-critical assumptions about reactor control loops and sensor bandwidth limits established through rigorous nuclear engineering standards [34], [35], [36]. Specifically, the following operational bounds are defined based on SMR control system specifications:

Maximum packet rate:

$$P_{\text{max}} = P_{\text{norm}} \left(1 + \frac{R_{\text{max}}}{100} \right), \quad (9)$$

where $P_{\text{norm}} = 1000$ packets/sec represents the nominal rate and $R_{\text{max}} = 500\%$ defines the allowable deviation.

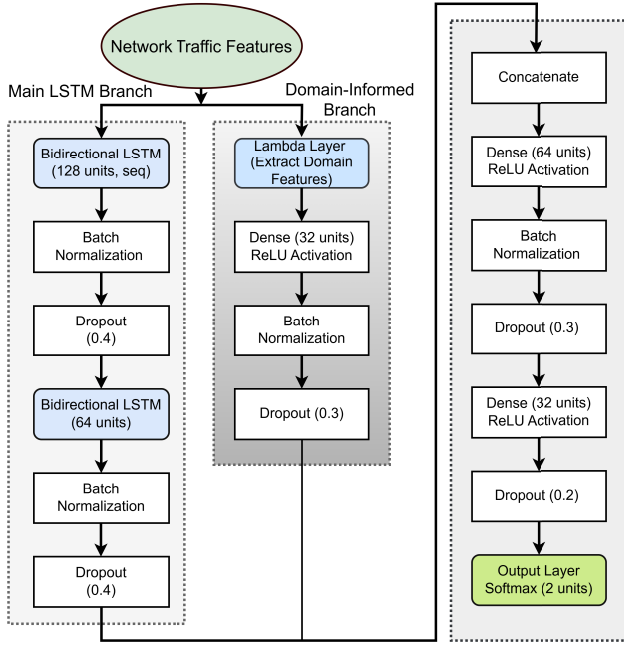


Fig. 2. Dual-branch Bi-LSTM model with domain-informed features for DDoS detection in SMR.

Maximum bandwidth:

$$B_{\max} = B_{\text{norm}} \left(1 + \frac{B_{\text{change}}}{100} \right), \quad (10)$$

with $B_{\text{norm}} = 10$ Mbps and $B_{\text{change}} = 800\%$.

Maximum packet size:

$$S_{\max} = 2 \cdot S_{\text{norm}}, \quad (11)$$

where $S_{\text{norm}} = 1500$ bytes.

Rate-of-change constraint on packet rate:

$$\left| \frac{dP(t)}{dt} \right| \leq \frac{P_{\max} - P_{\text{norm}}}{T_{\text{control}}}, \quad (12)$$

with $T_{\text{control}} = 100$ ms representing the response latency of the SMR control systems. These values serve as demonstration parameters for framework validation.

Due to the classified nature of actual SMR operational data, we employ representative parameter values for demonstration purposes, as our primary focus is to build the framework architecture. The domain-informed constraints are hypothesized and considered from established nuclear industry standards with parameters refined through empirical analysis of the CICIOT2023 dataset [37] to ensure operational relevance. The normal packet rate ($P_{\text{norm}} = 1000$ packets/sec) represents the typical traffic volume observed in industrial control networks used in nuclear facilities, which maintain constant message rates for sensor readings and control commands. The maximum allowable deviation ($R_{\max} = 500\%$) accounts for legitimate traffic spikes during reactor state transitions (e.g., startup sequences or emergency response procedures), while still identifying anomalous behavior beyond physical possibility. Similarly, the bandwidth parameters ($B_{\text{norm}} = 10$ Mbps, $B_{\text{change}} = 800\%$) reflect the bounded communication infrastructure in SMR environments, where network interfaces

are designed with specific capacity limits. The response time of the control system ($T_{\text{control}} = 100$ ms) is particularly significant, as it encodes the fundamental timing constraints of safety-critical control loops that cannot physically respond faster than their mechanical and electrical components allow [38], [39], [40], [41], [42].

To evaluate the adherence to operational constraints, we define a generalized domain compliance framework. Consider a cyber-physical system with M constraint dimensions. For each dimension k , we define a compliance function as $\varphi_k(x_k) = \min(1, x_k^{\text{bound}}/x_k)$. This function returns 1 when the observed value x_k stays within the operational bound x_k^{bound} . It degrades monotonically as the value exceeds the bound. The aggregate compliance score is the geometric mean across all dimensions: $\Phi(\mathbf{x}) = \left(\prod_{k=1}^M \varphi_k(x_k) \right)^{1/M}$. This ensures balanced penalization when any constraint is violated [43]. For SMR sensor networks, we instantiate our framework with $M = 3$ constraint dimensions using the operational bounds defined in Equations (9)–(11):

$$\varphi_{\text{packet}}(P) = \min \left(1, \frac{P_{\max}}{P} \right), \quad (13)$$

$$\varphi_{\text{bandwidth}}(B) = \min \left(1, \frac{B_{\max}}{B} \right), \quad (14)$$

$$\varphi_{\text{size}}(S) = \min \left(1, \frac{2 \cdot S_{\max}}{S} \right). \quad (15)$$

The aggregate domain compliance score is computed using the geometric mean to ensure balanced consideration of all physical dimensions:

$$\Phi(P, B, S) = (\varphi_{\text{packet}}(P) \cdot \varphi_{\text{bandwidth}}(B) \cdot \varphi_{\text{size}}(S))^{1/3}. \quad (16)$$

This formulation ensures a smooth penalty gradient and enforces proportional degradation for violations in any of the physical domain dimensions, guiding the model away from physically implausible network behavior patterns.

The dual-branch LSTM architecture is implemented through parallel processing steps. The main branch processes time-series network features through the Bi-LSTM stack described in the previous subsection. Concurrently, the domain-informed branch extracts and transforms physical features at the final time step, incorporating the following domain-specific attributes:

- Packet rate (packets/sec)
- Average packet size (bytes)
- Bandwidth usage (Mbps)
- Domain penalty score Φ

Let $\mathbf{X} \in \mathbb{R}^{T \times d}$ denote the input tensor, where T is the number of time steps and d is the number of features. The Domain-specific features are extracted using a custom Lambda function at the final time step T :

$$f_{\text{physics}}(\mathbf{X}) = \mathbf{X}_T, I_{\text{physics}}, \quad (17)$$

where I_{physics} indexes the physical domain features. These features are then processed through a nonlinear transformation pipeline:

$$\mathbf{z}_{\text{physics}} = \text{ReLU}(\mathbf{W}_{\text{physics}} f_{\text{physics}} + \mathbf{b}_{\text{physics}}), \quad (18)$$

$$\tilde{\mathbf{z}}_{\text{physics}} = \text{BatchNorm}(\mathbf{z}_{\text{physics}}), \quad (19)$$

$$\hat{\mathbf{z}}_{\text{physics}} = \text{Dropout}(\tilde{\mathbf{z}}_{\text{physics}}, p = 0.3). \quad (20)$$

The outputs from the LSTM temporal encoder $\mathbf{h}_{\text{LSTM}}$ and the domain-informed encoder $\hat{\mathbf{z}}_{\text{physics}}$ are concatenated to form a unified representation:

$$\mathbf{z}_{\text{combined}} = [\mathbf{h}_{\text{LSTM}}; \hat{\mathbf{z}}_{\text{physics}}]. \quad (21)$$

This joint embedding is passed through fully connected layers for final classification:

$$\mathbf{z}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{z}_{\text{combined}} + \mathbf{b}_1), \quad (22)$$

$$\tilde{\mathbf{z}}_1 = \text{BatchNorm}(\mathbf{z}_1), \quad (23)$$

$$\hat{\mathbf{z}}_1 = \text{Dropout}(\tilde{\mathbf{z}}_1, p = 0.3), \quad (24)$$

$$\mathbf{z}_2 = \text{ReLU}(\mathbf{W}_2 \hat{\mathbf{z}}_1 + \mathbf{b}_2), \quad (25)$$

$$\hat{\mathbf{z}}_2 = \text{Dropout}(\mathbf{z}_2, p = 0.2), \quad (26)$$

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{W}_3 \hat{\mathbf{z}}_2 + \mathbf{b}_3). \quad (27)$$

This dual-branch configuration ensures that the model not only learns abstract temporal representations but also incorporates network domain constraints into its decision boundary. During training, the inclusion of the penalty score Φ as a feature and regularizer guides the model away from physically implausible decisions, thus improving generalization and adversarial robustness in SMR-specific operational settings. However, deploying this model across distributed SMR sensor networks presents challenges related to data privacy, computational efficiency, and domain security separation. To address these challenges while preserving the benefits of the domain-informed approach, we integrate this architecture within a hierarchical FL framework, as detailed in the following subsection.

E. Federated Learning With Hierarchical Structure

The domain-informed dual-branch LSTM architecture provides a robust foundation for detecting DDoS attacks by incorporating both temporal network patterns and SMR-specific network domain constraints. However, deploying this model across distributed SMR sensor networks presents challenges related to data privacy, computational efficiency, and security domain separation. To preserve the confidentiality of sensor-level network traffic while enabling collaborative learning across SMR nodes, we implemented an FL architecture that allows decentralized nodes to train local models without exposing raw data, thus mitigating risks of centralized data leakage.

Not all nodes in an SMR environment exhibit equal trustworthiness or criticality due to their varying operational roles and security classifications. To account for this inherent heterogeneity, we employ a hierarchical FL structure that stratifies nodes into security tiers and applies weighted aggregation based on both tier level and trust score, as illustrated in Figure 3.

1) *Hierarchical Node Organization*: The Nodes are organized into three tiers according to their operational roles and associated security levels. This reflects the natural hierarchy present in SMR sensor networks [44]:

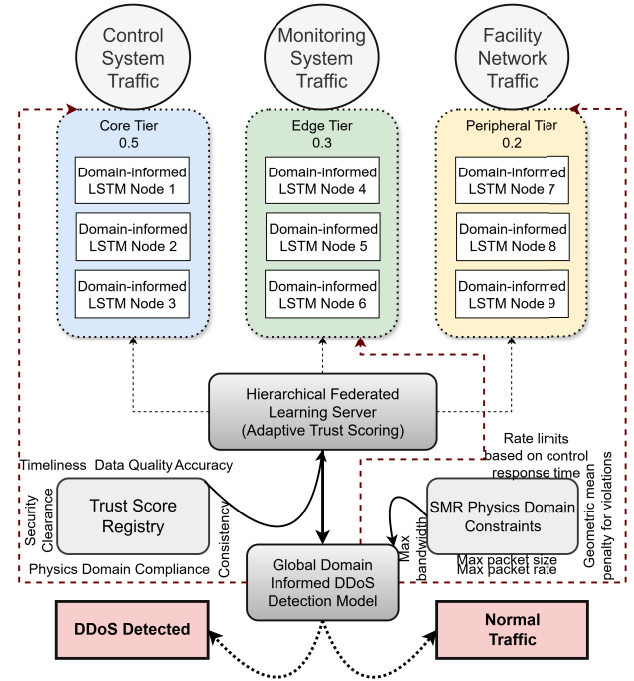


Fig. 3. Hierarchical FL with tier-aware aggregation across SMR nodes.

- **Core Tier**: Includes primary reactor control systems and critical infrastructure interfaces responsible for safety-critical operations such as reactor power regulation, coolant flow control, and emergency shutdown systems.
- **Edge Tier**: Encompasses secondary monitoring components, such as subsystems linked to local diagnostics, backup monitoring systems, steam generation controls, and non-critical control loops.
- **Peripheral Tier**: Represents ancillary facilities, general IT interfaces, communication systems, external interfaces, and less-trusted subsystems within the administrative building infrastructure.

Each tier is assigned an influence weight based on operational criticality and also can be varied ω_{tier} during model aggregation, reflecting its relative importance in the framework hierarchy:

$$\omega_{\text{tier}} = \begin{cases} 0.5, & \text{if tier} = \text{core}, \\ 0.3, & \text{if tier} = \text{edge}, \\ 0.2, & \text{if tier} = \text{peripheral}. \end{cases} \quad (28)$$

2) *Tier-Aware Node Selection Strategy*: For each FL round, a minimum number of nodes is selected from each tier to ensure comprehensive tier representation and maintain the hierarchical balance essential for SMR operational integrity:

$$|S_{\text{tier}}| \geq m_{\text{tier}}, \quad \forall \text{tier} \in \{\text{core}, \text{edge}, \text{peripheral}\}, \quad (29)$$

where S_{tier} is the set of selected nodes from the tier and m_{tier} is the minimum required nodes from each tier.

Additional nodes are selected based on available computational resources and tier weights to optimize the balance between representation and efficiency:

$$n_{\text{tier}} = \left\lceil \left(N_{\text{target}} - \sum_{\text{tier}} m_{\text{tier}} \right) \cdot \omega_{\text{tier}} \right\rceil, \quad (30)$$

where N_{target} denotes the total number of participants intended for a training round.

3) *Two-Level Aggregation Scheme*: The model aggregation process proceeds in two distinct stages to maintain both tier-specific characteristics and global model coherence:

Stage 1: Intra-Tier Aggregation - Within each tier, model updates are aggregated using trust scores τ_i and dataset sizes n_i to ensure that more reliable and data-rich nodes have greater influence within their tier:

$$\theta_{\text{tier}}^{t+1} = \frac{\sum_{i \in S_{\text{tier}}} \tau_i n_i \theta_i^{t+1}}{\sum_{i \in S_{\text{tier}}} \tau_i n_i}, \quad (31)$$

where θ_i^{t+1} denotes the model parameters submitted by node i after local training in round $t + 1$.

Stage 2: Inter-Tier Aggregation - Aggregated tier models are then combined into the global model based on the fixed tier weights, ensuring that core systems maintain strategic influence:

$$\theta_{\text{global}}^{t+1} = \sum_{\text{tier} \in \{\text{core}, \text{edge}, \text{peripheral}\}} \omega_{\text{tier}} \cdot \theta_{\text{tier}}^{t+1}. \quad (32)$$

If any tier lacks participants in a given round, its weight is redistributed proportionally across the remaining active tiers to maintain aggregation stability:

$$\omega_{\text{tier}}^{\text{adjusted}} = \frac{\omega_{\text{tier}}}{\sum_{\text{tier}' \in \text{active tiers}} \omega_{\text{tier}'}}. \quad (33)$$

4) *Implementation Details*: The hierarchical structure is realized through a multi-class node registration and routing system that dynamically allocates clients to tiers based on their operational roles and security classifications. In each FL round, the server utilizes both the node's current trust score and its tier affiliation to determine participation eligibility, ensuring that only qualified nodes contribute to model updates.

Aggregation is performed by custom routines that normalize updates according to Equation (31) and apply tier weights via Equation (32). This implementation directly corresponds to the `HierarchicalFederatedLearning` class in our codebase, which manages tier assignments, node selection, and hierarchical aggregation through the `select_nodes_for_round()` and `aggregate_models()` methods.

This tier-aware aggregation scheme offers several advantages for SMR environments. Firstly, it reflects the operational hierarchy inherent in SMRs by giving more influence to nodes with higher safety criticality, ensuring that critical reactor control systems maintain primary decision-making authority. Secondly, it mitigates the risk posed by compromised or untrusted nodes in lower tiers, preventing malicious updates from peripheral systems from affecting the global model. Finally, it aligns with defense-in-depth principles by embedding institutional trust boundaries directly into the learning process, creating multiple layers of security that correspond to the physical security zones within SMR facilities.

The hierarchical FL framework thus provides a foundation for secure, distributed learning that respects the operational and security requirements of SMR environments while enabling collaborative DDoS detection across diverse network

components. Building upon this hierarchical structure, the next subsection details the adaptive trust scoring mechanism that dynamically evaluates node reliability and guides participation decisions.

F. Adaptive Multi-Dimensional Trust Scoring

Following the establishment of the hierarchical FL structure, robust FL in SMR environments demands accurate evaluation of participating nodes, particularly when they differ in behavior, reliability, or susceptibility to cyber compromise. The heterogeneous nature of the sensor data comprising core reactor control systems, edge monitoring components, and peripheral facility devices data requires a sophisticated trust assessment mechanism that can dynamically adapt to varying node performance and operational conditions. To capture this heterogeneity and ensure reliable model aggregation, we design an adaptive trust scoring mechanism that evaluates each node based on six orthogonal dimensions. The resulting trust score τ_i informs both the selection of nodes and the aggregation of the model, ensuring that the most trustworthy nodes exert greater influence on the global model while mitigating the risks of compromised or unreliable participants.

1) *Multi-Dimensional Evaluation Criteria*: Each node i in the SMR federated network is evaluated across the following six dimensions, selected to capture both technical performance and operational trustworthiness:

- d_1 : **Accuracy** — Predictive performance of the local model measured through validation metrics.
- d_2 : **Data Quality** — Integrity and representativeness of the node's dataset, including completeness and noise levels.
- d_3 : **Timeliness** — Responsiveness in submitting local updates within prescribed federated learning rounds.
- d_4 : **Consistency** — Stability of performance over recent rounds, indicating operational reliability.
- d_5 : **Domain Compliance** — Adherence to SMR-specific physical network domain constraints as defined in the previous subsection.
- d_6 : **Security Clearance** — Node's baseline trust level based on tier classification or security access level.

Each dimension d is assigned a weight ω_d reflecting its relative importance in the SMR operational context:

$$\omega_d = \begin{cases} 0.25 & \text{if } d = d_1 \text{ (accuracy),} \\ 0.20 & \text{if } d = d_2 \text{ (data quality),} \\ 0.15 & \text{if } d \in \{d_3, d_4, d_5\}, \\ 0.10 & \text{if } d = d_6 \text{ (security clearance).} \end{cases} \quad (34)$$

The total trust score is computed as a weighted sum across all dimensions:

$$\tau_i = \sum_{d=1}^6 \omega_d \cdot s_{i,d}, \quad (35)$$

where $s_{i,d}$ denotes the score of node i in dimension d , normalized to the interval $[0, 1]$.

2) *Domain Compliance Scoring*: To integrate domain knowledge into trust estimation and maintain consistency with

the domain-informed architecture, we incorporate the domain penalty function defined in Equation (16). For node i with local dataset $\mathcal{D}_i$, the domain compliance score is calculated as:

$$s_{i,d_5} = \frac{1}{|\mathcal{D}_i|} \sum_{x \in \mathcal{D}_i} \Phi(P_x, B_x, S_x), \quad (36)$$

ensuring that nodes transmitting physically implausible traffic patterns are penalized, thereby aligning trust assessment with SMR operational integrity and the domain-informed detection principles established in the dual-branch architecture.

3) *Adaptive Trust Updates*: Trust scores are dynamically updated after each FL round using exponential smoothing to incorporate recent performance while maintaining historical context:

$$s_{i,d}^{t+1} = \alpha_d \cdot v_{i,d}^t + (1 - \alpha_d) \cdot s_{i,d}^t, \quad (37)$$

where $v_{i,d}^t$ is the observed metric in round t and α_d is a dimension-specific smoothing factor. For domain compliance, we apply heavier penalization for severe violations to maintain safety-critical operational boundaries:

$$\alpha_{d_5} = \begin{cases} 0.5 & \text{if } v_{i,d_5}^t < 0.5, \\ 0.3 & \text{otherwise.} \end{cases} \quad (38)$$

4) *Consistency Evaluation*: To capture temporal stability and identify erratic behavior that can indicate compromised or unreliable nodes, the consistency score is derived from the variance of historical performance across accuracy and data quality dimensions. Given a decay factor $\gamma = 0.9$, the weighted variance is computed as:

$$\text{Var}_w(s_{1:t,i,d'}) = \frac{\sum_{j=1}^t \gamma^{t-j} (s_{j,i,d'} - \mu_w)^2}{\sum_{j=1}^t \gamma^{t-j}}, \quad (39)$$

where the exponentially weighted mean is:

$$\mu_w = \frac{\sum_{j=1}^t \gamma^{t-j} s_{j,i,d'}}{\sum_{j=1}^t \gamma^{t-j}}. \quad (40)$$

The consistency score is then calculated as:

$$s_{i,d_4}^t = 1 - 10 \cdot \frac{\sum_{d' \in \{d_1, d_2\}} \text{Var}_w(s_{1:t,i,d'})}{2}. \quad (41)$$

This formulation penalizes erratic behavior over time, enabling the identification of unstable or potentially compromised nodes that exhibit inconsistent performance patterns.

5) *Integration and Export*: The trust module maintains a dynamic trust map across all participating nodes and exports the scores for use in both the hierarchical aggregation process and secure audit trails. By coupling data-driven performance metrics with domain-aware scoring, this mechanism provides a comprehensive and context-sensitive assessment of trustworthiness specifically tailored for SMR federated environments.

The adaptive nature of this trust scoring function ensures that nodes demonstrating consistent high performance and domain compliance receive greater influence in model aggregation, while nodes exhibiting suspicious behavior or poor performance are appropriately down-weighted or excluded from critical aggregation rounds. This approach maintains the

integrity of the global detection model while accommodating the dynamic and heterogeneous nature of SMR sensor networks, thereby establishing the foundation for secure aggregation and framework integration, as detailed in the following subsection.

G. Secure Aggregation and System Integration

Our architectural approach brings together several critical components. We integrate temporal learning capabilities with physical network domain expertise. Additionally, we incorporate adaptive trust scoring mechanisms. All of these elements operate within a privacy-preserving federated framework. This comprehensive integration creates a robust defense strategy. The strategy specifically targets the unique challenges found in SMR cyber-physical systems. These systems operate in mission-critical environments where security cannot be compromised. Protecting model updates during transmission presents significant challenges. We address this through secure aggregation protocols. These protocols must align with strict SMR regulatory requirements [45]. They also support the defense-in-depth strategies that nuclear infrastructure protection demands [46]. The model update structure we have developed serves multiple purposes. Updates flow from clients to our hierarchical FL server through carefully designed channels. This structure enables encrypted aggregation while maintaining tier-specific weighting capabilities. Importantly, it does this without exposing individual contributions or sensitive operational data.

The integrated FL workflow proceeds through the following systematic phases:

Phase 1: Local Model Training and Domain Compliance Assessment

Each distributed SMR node develops its local dual-branch DI-LSTM model through training on private datasets. This process follows the architecture outlined in Subsections III-B and III-C. The nodes simultaneously compute compliance penalties for the network domain using Equation (16), which ensures that the local models naturally adhere to the operating boundaries of the SMR. The domain penalty function $\Phi(P, B, S)$ operates dual roles—it functions as both feature input and regularization tool. This design steers local models from physically impossible decisions while preserving their ability to detect legitimate attack signatures.

Phase 2: Secure Parameter Transmission and Trust Evaluation

The distributed nodes forward their updated model parameters $\theta_i^{(t+1)}$ to the central aggregator via secure communication pathways. Along with these parameters, trust metrics travel through the same channels, including accuracy measurements, domain compliance scores, and data quality indicators. Participation metadata also accompany this transmission. The server conducts thorough evaluation of multi-dimensional trust scores through its adaptive assessment framework. This process utilizes the scheme detailed in Equations (35)–(41). Real-time performance data merge with historical consistency records during this evaluation. This integration creates a comprehensive approach to the robust assessment of the reliability of the nodes.

Phase 3: Hierarchical Node Selection and Tier-aware Aggregation

The FL aggregation server selects qualified node subsets from each operational tier through its tier-aware selection approach. This strategy ensures balanced representation spans core, edge, and peripheral systems. Priority goes to nodes demonstrating higher trust scores along with the best network domain compliance ratings. The selection framework preserves SMR's natural operational hierarchy. Moreover, it blocks malicious or unreliable nodes from undermining the integrity of the global model.

Phase 4: Two-Level Model Aggregation

Individual tier models undergo aggregation through trust-weighted averaging, following the specifications in Equation (31). Inter-tier aggregation follows this process, applying fixed tier weights according to Equation (32). This hierarchical approach preserves the strategic influence of critical reactor control nodes within the global model.

Phase 5: Global Model Distribution and Continuous Adaptation

The aggregated global model $\theta_{\text{global}}^{(t+1)}$ is distributed back to all participating nodes. Each node replaces its local model parameters with the updated global parameters before the next training round begins. The domain-informed constraints defined in Equation (16) remain enforced during subsequent local training, and trust scores are carried forward to inform node selection in the next round. This cycle repeats for T communication rounds until convergence.

H. Theoretical Convergence Analysis

To provide theoretical analysis for the faster convergence observed in our framework, we analyze how the hierarchical tier-weighted aggregation and adaptive trust scoring jointly reduce the effective variance of the global model updates.

In standard FL with uniform aggregation, the global model update at round t is given by $\theta^{(t+1)} = \frac{1}{K} \sum_{i=1}^K \theta_i^{(t+1)}$, where K is the number of participating clients. Under non-IID data distributions, the variance of the aggregated update is bounded by [47]:

$$\mathbb{E} \left[\left\| \theta^{(t+1)} - \theta^* \right\|^2 \right] \leq \frac{1}{K} \sum_{i=1}^K \sigma_i^2 + \Gamma^2, \quad (42)$$

where σ_i^2 denotes the local gradient variance at node i and Γ^2 captures the divergence due to non-IID data heterogeneity across clients.

In our framework, the global update follows two-level weighted aggregation. At the intra-tier level, updates are weighted by trust scores τ_i (Equation (31)), and at the inter-tier level, tier weights ω_{tier} are applied (Equation (32)). Our DHFL aggregation can be expressed as:

$$\theta_{\text{DHFL}}^{(t+1)} = \sum_{\text{tier}} \omega_{\text{tier}} \cdot \frac{\sum_{i \in S_{\text{tier}}} \tau_i \theta_i^{(t+1)}}{\sum_{i \in S_{\text{tier}}} \tau_i}. \quad (43)$$

We now show that this formulation reduces the effective update variance. Let $w_i = \omega_{\text{tier}(i)} \cdot \tau_i / \sum_{j \in S_{\text{tier}(i)}} \tau_j$ denote the

effective weight assigned to client i in our framework. The variance of the weighted aggregation follows:

$$\text{Var}(\theta_{\text{DHFL}}^{(t+1)}) = \sum_i w_i^2 \cdot \sigma_i^2 \leq \left(\max_i w_i \right) \cdot \sum_i w_i \cdot \sigma_i^2. \quad (44)$$

Since the trust scoring mechanism assigns higher weights τ_i to nodes with lower local variance (i.e., more consistent and accurate clients) and lower weights to unreliable nodes, the trust-weighted scheme follows:

$$\sum_i w_i \cdot \sigma_i^2 \leq \frac{1}{K} \sum_{i=1}^K \sigma_i^2, \quad (45)$$

This inequality holds under the assumption that the adaptive trust scoring mechanism (Equations (35)–(41)) assigns higher aggregate scores to nodes exhibiting lower local gradient variance, which is supported by the inclusion of accuracy (d_1) and consistency (d_4) as dominant trust dimensions with combined weight $\omega_{d_1} + \omega_{d_4} = 0.40$ in Equation (34).

Furthermore, the hierarchical structure contributes to variance reduction by partitioning clients into tiers with distinct data characteristics. The inter-tier aggregation with fixed weights ω_{tier} acts as a stratified sampling estimator. By the law of total variance:

$$\text{Var}(\theta_{\text{DHFL}}) = \sum_{\text{tier}} \omega_{\text{tier}}^2 \cdot \text{Var}_{\text{intra}}(\text{tier}) + \text{Var}_{\text{inter}}(\bar{\theta}_{\text{tier}}), \quad (46)$$

where the first term captures within-tier variance (reduced by trust weighting) and the second term captures between-tier variance (controlled by the fixed tier weights). When the tier weights ω_{tier} are chosen proportional to the operational reliability of each tier, as in our framework where $\omega_{\text{core}} = 0.5 > \omega_{\text{edge}} = 0.3 > \omega_{\text{peripheral}} = 0.2$, the inter-tier variance component is minimized by giving greater influence to the most stable tier (core), which processes higher-quality data with lower gradient noise.

By combining these two mechanisms, we obtain the following convergence bound for our framework after T communication rounds:

$$\mathbb{E} \left[\left\| \theta_{\text{DHFL}}^{(T)} - \theta^* \right\|^2 \right] \leq \frac{\eta}{T} \left(\sum_{\text{tier}} \omega_{\text{tier}}^2 \cdot \bar{\sigma}_{\text{tier}}^2 \right) + \Gamma_{\text{DHFL}}^2, \quad (47)$$

where $\bar{\sigma}_{\text{tier}}^2$ is the trust-weighted average variance within each tier and Γ_{DHFL}^2 represents the residual heterogeneity after hierarchical stratification. Since trust weighting ensures $\bar{\sigma}_{\text{tier}}^2 \leq \bar{\sigma}_{\text{uniform}}^2$ (Inequality (45)) and hierarchical stratification reduces $\Gamma_{\text{DHFL}}^2 \leq \Gamma^2$ by grouping nodes with similar operational profiles, our framework convergence bound in Equation (47) is strictly tighter than the standard FedAvg bound in Equation (42). This theoretical result explains the empirical observation that our framework converges within 30–50 rounds compared to baseline methods.

IV. PERFORMANCE EVALUATION

The experimental evaluation is being done in a controlled cloud-based virtual environment that simulates the distributed nature of SMR IoT sensor networks. This environment mimics real-world deployment scenarios with multiple FL clients

TABLE II
HARDWARE AND SOFTWARE SPECIFICATION OF THE
SIMULATION ENVIRONMENT

Component	Description
Operating System	Windows 11
Processor	Intel® Core™ Ultra 5 235 vPro
Random Access Memory (RAM)	32 GB
Graphic Cards	NVIDIA Tesla T4/V100 GPU
Environment	Cloud-based virtual environment
FL Framework	Flower (flwr) 1.1.0 TensorFlow 2.15.0
Deep Learning Library	TensorFlow 2.15.0 with Keras API
Scientific Computing	NumPy 1.25.2, Pandas 2.0.3
Machine Learning Library	Scikit-learn 1.3.0
Visualization	Matplotlib 3.7.1, Seaborn 0.12.2
Network Simulation	30 distributed clients
Dataset	CICIoT2023 (2.8 million records)
Communication Rounds	Up to 50 FL rounds

arranged in hierarchical tiers. Each tier represents different levels of computational power and network limitations, similar to what is found in actual nuclear facility infrastructures. The framework supports up to 30 distributed FL clients and allows flexible configuration of communication rounds, data splits, and training behaviors. The use of GPUs such as NVIDIA Tesla T4 or V100 accelerates training, while modern libraries like TensorFlow, NumPy, and Scikit-learn support DL and data processing tasks. Flower (flwr) 1.1.0 manages the FL orchestration, enabling up to 50 communication rounds. The CICIoT2023 dataset, which includes 2.8 million records, helps model realistic attack scenarios and normal traffic patterns. This ensures that the simulation reflects the complexity of real-world SMR networks. Table II summarizes the hardware and software used in the simulation environment.

A. Dataset Description

We evaluated our DHFL framework using the CICIoT2023 dataset [37], which contains approximately 464 million IoT network flows and 33 attack types, including DDoS variants such as HTTP, UDP and TCP floods, as well as Slowloris. These attacks align closely with the threat scenarios faced by SMR environments, such as the 2022 Energoatom incident [6], where nuclear infrastructure was targeted by sophisticated DDoS attacks. Due to the classified nature of real SMR operational data (as outlined in our problem formulation) and to ensure computational efficiency and balanced representation across attack types, we strategically sample 2.8 million records from the full CICIoT2023 dataset. This subset preserves the original class distribution, maintaining a realistic imbalance of approximately 78% normal traffic and 22% DDoS attack traffic. CICIoT2023 is a representative dataset for our study, as its IoT architecture mirrors our proposed three-tier SMR structure. We retain the non-IID distribution constraint, where the ratio $|DDoS|_i/|Normal|_i$ varies across tiers, allowing us to evaluate the robustness of our `HierarchicalFederatedLearning` class in handling tier-specific data allocation. Our main objective is to demonstrate the architecture of the framework and the effectiveness of the hierarchical FL approach using a representative subset that captures key DDoS attack patterns relevant to SMR security.

For federated simulation, the dataset is partitioned across 30 nodes according to our hierarchical structure with

non-IID distributions. The core nodes, which represent critical reactor control systems, receive 40% of high-quality data to reflect their operational importance, with a tier weight of $\omega_{core} = 0.5$. Edge nodes, which simulate secondary monitoring systems, are allocated 35% of moderately variable data with a weight of $\omega_{edge} = 0.3$. The peripheral nodes, corresponding to larger facility networks, handle 25% of higher-noise patterns with $\omega_{peripheral} = 0.2$. This distribution is implemented in our `create_federated_datasets()` function and is enforced through the tier weights defined in Equation (28). To assess the compliance of each node with the operational constraints of the SMR, we pre-calculate the domain compliance scores $\Phi(P, B, S)$ using a geometric mean penalty function. These scores are integrated into our `AdaptiveTrustScorer` class, which enables the framework to distinguish between legitimate operational transients of the reactor (e.g., initialization sequences) and genuine cyberattacks. This domain-informed detection is supported by our dual-branch architecture, which processes both temporal patterns and domain-specific network constraints simultaneously.

B. Implementation Details and Hyperparameters

Our FL simulation implements a hierarchical framework that mirrors real-world SMR deployment scenarios through systematic evaluation phases. The simulation orchestrates 30 distributed clients across three operational tiers: 5 core nodes (representing critical reactor control systems), 10 edge nodes (secondary monitoring components) and 15 peripheral nodes (facility networks), conducting up to 50 communication rounds to ensure convergence stability. Each federated round follows a structured workflow. It begins with local domain-informed dual-branch Bi-LSTM training that incorporates network constraints. This is followed by secure parameter transmission and multi-dimensional trust evaluation across six dimensions: accuracy, data quality, timeliness, consistency, domain compliance, and security clearance. Hierarchical node selection is applied to ensure minimum representation of each tier. The aggregation process includes two levels: trust-weighted intra-tier averaging, followed by tier-weighted inter-tier combination using fixed weights ($\omega_{core} = 0.5$, $\omega_{edge} = 0.3$, $\omega_{peripheral} = 0.2$). The simulation incorporates a non-IID data distribution, where different tiers experience varying attack patterns. The core nodes receive 40% of high-quality data to reflect their operational criticality, the edge nodes obtain 35% with moderate variability, and the peripheral nodes handle 25% with higher noise levels. Our adaptive trust scoring mechanism dynamically evaluates the reliability of the nodes using exponential smoothing with dimension-specific weights. Furthermore, domain compliance scoring integrates the geometric mean penalty function $\Phi(P, B, S)$ to enforce compliance to SMR-specific operational constraints throughout the FL process.

C. Results and Analysis

Figure 4(a) shows the overall learning progress of the global model in terms of accuracy and F1-score over 50 communication rounds. The accuracy increases steadily from around 68%

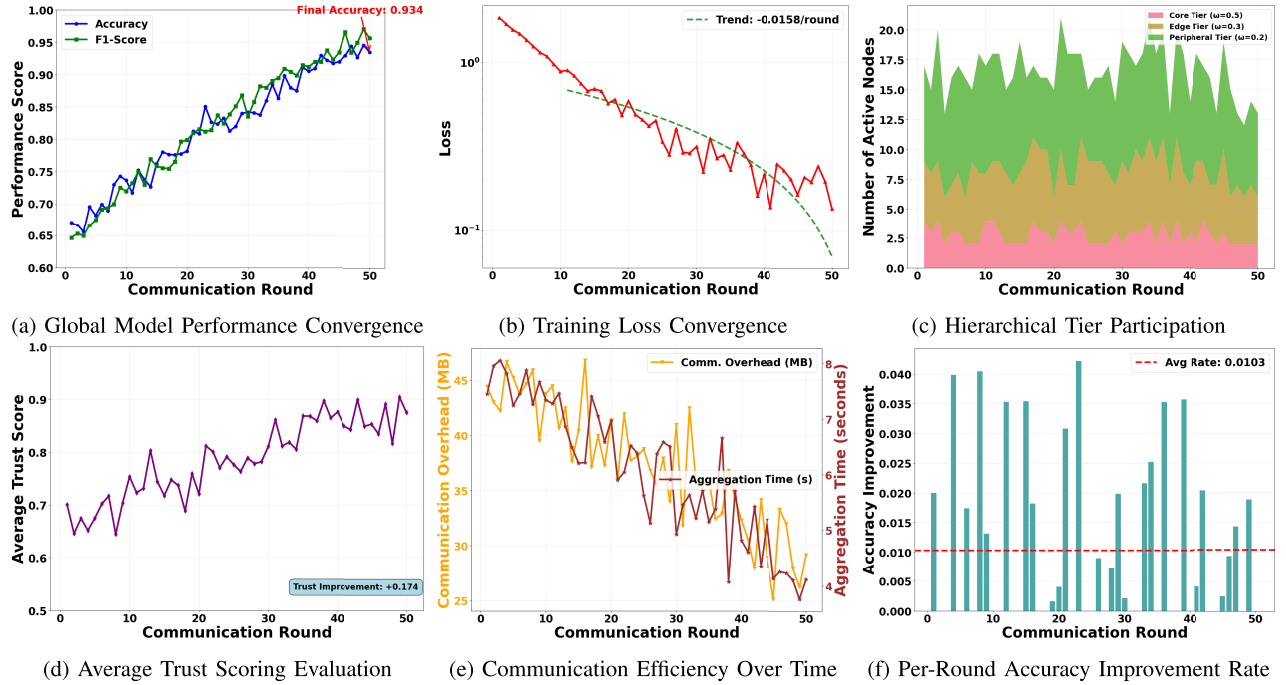


Fig. 4. Performance metrics of various FL parameters for proposed DHFL framework.

in the first few rounds to 93.4% in the final round, showing a total improvement of approximately 25.4 percentage points. The F1-score follows a similar trend, reaching a final value of 95.5%, indicating an excellent precision-recall balance. This strong convergence behavior confirms that the federated learning process effectively learns from non-IID data across all tiers. The absence of major fluctuations shows that the framework is stable and resistant to low-quality or noisy client updates. Figure 4(b) shows a clear downward trend in training loss, following an exponential decay pattern with a slope of approximately -0.015 per round. Initially, the loss starts above 1.0, but at the end of the simulation, it drops below 0.1, marking a 90% reduction. This consistent decrease in loss suggests that the model progressively minimizes its error, even when aggregating knowledge from heterogeneous clients.

Figure 4(c) shows how many nodes from each tier actively participate in each communication round. The core nodes with weight ($\omega = 0.5$) contribute 2 to 4 nodes per round, the edge nodes ($\omega = 0.3$) contribute 4 to 7 nodes, and the peripheral nodes ($\omega = 0.2$) contribute 6 to 11 nodes. Consistent participation from all tiers helps the model learn from different types of data and system roles, which is important in SMR environments where both core and peripheral systems matter. The participation distribution also validates our hierarchical client selection policy, which maintains proportional tier-wise representation in every round. The graph in Figure 4(d) presents the evolution of the average trust scores over time, starting at 0.65 and increasing to 0.87 by the final round, showing an improvement in trust scores of 0.174 or approximately 26.7% gain. This increase demonstrates the effectiveness of our adaptive trust evaluation system that monitors six critical dimensions (accuracy, data quality, timeliness, consistency, domain compliance, and security clearance). Over time, less reliable or inconsistent clients are excluded or

contribute less to the global model, while high-trust nodes are favored, resulting in stronger convergence and improved model robustness.

Figure 4(e) shows two critical metrics: communication overhead in megabytes (MB) and aggregation time in seconds. Initially, the communication overhead is about 45MB, which gradually decreases to around 30MB, a 33% reduction over the course of training. Similarly, the aggregation time decreases from approximately 8 seconds to 4 seconds, also indicating a 50% improvement. These results highlight the benefits of our tier-aware aggregation strategy and selective client participation, which reduce the communication burden and make the framework more scalable and practical for real-world SMR deployments. Figure 4(f) shows the improvement in the accuracy of the model in each round. While the improvement is not uniform across all rounds, the average per-round improvement is 0.0103, or roughly 1.03% per round. Some rounds show sharp increases of up to 0.042, while others offer minimal gains. The variation indicates that certain rounds contribute significantly more, often when high-trust nodes of the core or edge tiers are involved.

Figure 5 shows the confusion matrix analysis, which reveals the steady detection capabilities of the framework. The model correctly classifies 23,718 normal instances and 67,988 DDoS attacks, while only 2,309 normal samples are misclassified as DDoS and 1,690 DDoS samples are misclassified as normal. This shows a high true positive rate for DDoS detection and a low false alarm rate. These results confirm our hypothesis that the integration of SMR-specific domain constraints through the penalty function $\Phi(P, B, S)$ within the hierarchical FL process significantly improves both the detection accuracy and the reliability of the framework. By aligning the learning process with the operational context of nuclear systems, the framework not only improves classification performance but

TABLE III
PERFORMANCE COMPARISON WITH EXISTING FEDERATED LEARNING METHODS

FL Method	Accuracy	Precision	Recall	F1-Score	AUC	Target System
[19]	0.887	0.882	0.891	0.886	0.912	Wireless IoT with UAV
[20]	0.923	0.918	0.928	0.923	0.946	Healthcare IoMT
[28]	0.901	0.896	0.907	0.901	0.925	IoT Non-IID Data
[29]	0.889	0.885	0.893	0.889	0.914	Precision Agriculture
DHFL	0.934	0.945	0.975	0.955	0.989	SMR Sensor

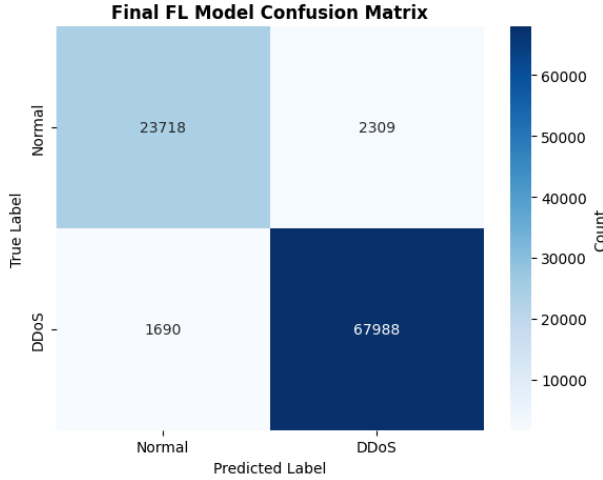


Fig. 5. Confusion matrix of the proposed DHFL framework.

also supports the cybersecurity requirements of critical SMR infrastructure.

D. Ablation Studies

To evaluate individual contribution of each of our framework's components, we conduct a systematic ablation study by selectively disabling one component at a time while keeping all other elements unchanged. Specifically, we evaluate three ablated variants against the full DHFL configuration: (i) *DHFL without (w/o) Physics*, which removes the domain-informed dual-branch architecture and trains a standard Bi-LSTM without the physics penalty function $\Phi(P, B, S)$; (ii) *DHFL w/o Hierarchy*, which replaces the tier-weighted aggregation with uniform weights and assigns all nodes to a single tier, effectively reducing the framework to flat FedAvg; and (iii) *DHFL w/o Trust*, which fixes all trust scores at $\tau_i = 1.0$ and disables the adaptive multi-dimensional trust update mechanism, treating all participating nodes as equally reliable. Each variant is trained under identical conditions using the same CICIoT2023 dataset, 30 federated clients, non-IID data distribution, and 50 communication rounds.

In Table V, we present the detection performance of each configuration. The ablation results demonstrate the impact of each component in our DHFL framework. Removing the physics constraints leads to a drop in precision and AUC, which indicates that the domain-informed penalty function $\Phi(P, B, S)$ is making an impact for reducing false positives

by enforcing physically plausible behavior. Eliminating the hierarchical aggregation causes the largest decline in accuracy and overall performance, which highlights its critical role in handling non-IID data and preserving the influence of high-importance nodes. In contrast, removing the trust scoring mechanism results in a moderate degradation across all metrics, showing that trust-aware client selection improves robustness by limiting the effect of unreliable updates. Finally, the full DHFL framework achieves the highest accuracy of 93.4% with an AUC of 0.989, confirming that all three components contribute positively to the overall detection capability.

E. Comparative Analysis With State-of-the-Art Methods

The experimental evaluation shows that our proposed framework achieves consistently strong results across multiple FL metrics. As shown in Table III, our framework achieves an accuracy of 93.4%, precision of 94.5%, recall of 97.5%, F1-score of 95.5%, and AUC of 98.9%. Compared to the best-performing baseline method [20], which records 92.3% accuracy and 92.3% F1-score, our framework shows an overall improvement of approximately 1 to 3% across key performance metrics such as accuracy, F1-score, and recall. These gains are especially significant for SMR sensor networks, where high recall (i.e., minimizing false negatives) is critical to detect DDoS attacks that could otherwise impact operational safety.

In terms of communication efficiency, Table IV shows that our framework reaches convergence within 30 to 50 communication rounds, compared to 60 to 110 rounds required by other methods, a reduction of up to 55% in training rounds. This is made possible by our adaptive trust scoring mechanism, which evaluates clients in six dimensions: accuracy, data quality, timeliness, consistency, domain compliance, and security clearance, and selects high-trust clients for participation. This approach enables faster convergence, reduces communication overhead, and supports both privacy and scalability requirements, making it well-suited for deployment in resource-constrained and mission-critical SMR infrastructures.

Fig. 6 shows a comparison of the convergence speed between our proposed framework and other existing federated learning methods. The results show that our framework achieves an accuracy greater than 90% in just 32 communication rounds, which is faster than other approaches such as UAV-FL (75 rounds), Healthcare FL (55 rounds), Non-IID FL (68 rounds) and Agriculture FL (82 rounds). This translates to a 40% to 60% reduction in training rounds,

TABLE IV
FEDERATED LEARNING COMMUNICATION AND COMPUTATIONAL EFFICIENCY COMPARISON

FL Method	Communication Rounds	Convergence Speed	Client Selection	Aggregation Method	Privacy Level	Scalability Support
[19]	80–100	Medium	Random	FedAvg	Basic	Medium
[20]	60–80	Fast	Tier-based	Weighted	High	High
[28]	70–90	Medium	Gale-Shapley	Enhanced	Medium	High
[29]	85–110	Slow	Geographic	Personalized	High	Medium
[48]	40–60	Fast	Dominant Class-based	Multi-teacher KD + Contrastive Learning (Edge)	Medium	High
DHFL	30–50	Fast	Trust+Tier	Hierarchical	High	High

TABLE V

ABLATION STUDY: COMPONENT CONTRIBUTION ANALYSIS (P = PHYSICS CONSTRAINTS, H = HIERARCHICAL AGGREGATION, T = TRUST SCORING)

Configuration	Components	Accuracy	AUC	F1-Score	Precision	Recall
Full DHFL	P+H+T	0.934	0.989	0.955	0.945	0.975
w/o Physics	H+T	0.917	0.956	0.934	0.905	0.966
w/o Hierarchy	P+T	0.878	0.946	0.910	0.863	0.963
w/o Trust	P+H	0.922	0.955	0.934	0.916	0.954

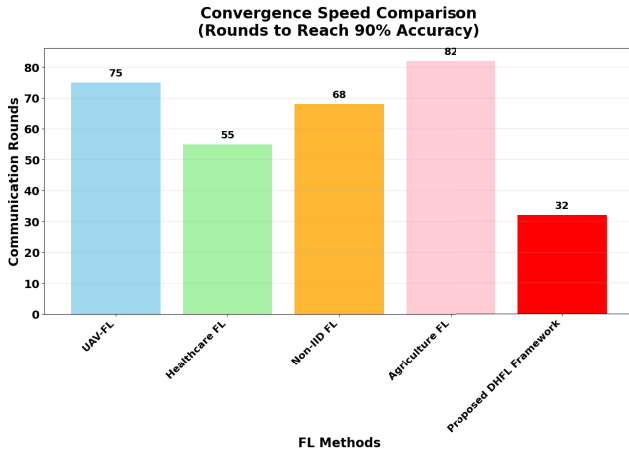


Fig. 6. Convergence speed comparison with state-of-the-art methods.

depending on the baseline, and highlights the effectiveness of our trust- and tier-aware aggregation strategy in accelerating learning. In addition to faster convergence, our framework also demonstrates enhanced communication efficiency. The total communication cost is limited to 1,480 MB, reflecting both the reduced number of rounds and the selective participation of high-trust nodes. By minimizing redundant or low-quality updates, the framework ensures that only meaningful contributions are exchanged and aggregated. This not only reduces bandwidth usage, but also shortens training time, making the solution more scalable and suitable for resource-constrained environments. By addressing slow convergence, high communication cost, and lack of domain trust, our framework provides a practical and efficient solution to secure learning in nuclear environments.

F. Cross-Dataset Validation: A Case Study With CIC-DDoS2019

To assess the generalizability of our framework beyond the primary evaluation dataset, we conduct a cross-dataset validation using the CIC-DDoS2019 dataset [49]. The framework achieves 87.5% accuracy, 91.7% AUC, and 90.3% F1 score on CIC-DDoS2019. While these numbers are lower than the

CICIoT2023 results, the performance gap is expected because the two datasets have fundamentally different characteristics in terms of network topology, attack types, and traffic generation tools. Additionally, the feature distributions differ substantially between the two datasets. Although both share flow-based features, their statistical properties vary due to different network configurations. Despite these differences, all three framework components, including the hierarchical aggregation, trust scoring, and domain penalty function, operate on this unseen dataset.

G. Discussion

Our experimental results, presented in this section, demonstrate the effectiveness of our framework across multiple evaluation dimensions. However, we acknowledge several considerations that contextualize these findings.

First, the experimental validation relies on publicly available IoT datasets rather than real SMR operational data. This is a practical constraint shared across the nuclear cybersecurity research community, as actual SMR network traffic data is classified for national security reasons. No publicly available dataset currently captures real SMR sensor network communications. Recent SMR cybersecurity studies face the same limitation: Salehpour et al. [13] used RTDS simulation without domain-informed or federated learning components, and Ayo-deji et al. [30] relied on the Asherah Nuclear Simulator without privacy-preserving mechanisms. We selected CICIoT2023 as our primary dataset because its hierarchical IoT architecture closely mirrors our proposed three-tier SMR structure, and it contains DDoS attack variants that align with real-world nuclear infrastructure threats such as the 2022 Energoatom incident [6]. The cross-dataset validation with CIC-DDoS2019 dataset further confirms that our framework is not tailored to a single benchmark.

Second, the primary contribution of our work is the framework architecture itself. The generalized domain compliance framework introduced in Subsection III-D is designed so that domain expert can instantiate it by specifying operational bounds for their specific system. The theoretical convergence analysis (Subsection III-H) and the ablation study

(Subsection IV-D) validate the design choices independently of any particular dataset. This positions our framework as a reusable and adaptable solution for federated DDoS detection in hierarchical sensor networks across different critical infrastructure domains.

Third, future work will focus on two key directions: (i) deploying the framework in a simulated SMR testbed to assess practical scalability and regulatory compliance, and (ii) expanding the detection scope beyond DDoS attacks to encompass more sophisticated cyber threats.

V. CONCLUSION

This paper presented a Domain-informed Hierarchical Federated Learning (DHFL) framework for detecting DDoS attacks in SMR IoT sensor networks. The framework integrated a dual-branch Bi-LSTM architecture with network-specific domain constraints and a multi-dimensional adaptive trust mechanism, enabling accurate and privacy-preserving collaborative learning across core, edge, and peripheral tiers. Through tier-weighted aggregation and domain penalty enforcement, the framework maintained SMR operational integrity while improving detection accuracy. The experimental evaluation demonstrated robust performance with 93.4% accuracy, 97.5% recall, and 98.9% AUC, along with faster convergence and lower communication overhead compared to existing methods. These results suggest the suitability of the framework for deployment in safety-critical IoT environments where privacy, reliability, and system-specific compliance are essential. Future work will focus on expanding the framework beyond DDoS attacks to encompass sophisticated cyber threats, such as false data injection and persistent threats, and deploying the framework in a simulated testbed to assess practical scalability and regulatory compliance.

REFERENCES

- [1] H. F. Rashvand, A. Abedi, J. M. Alcaraz-Calero, P. D. Mitchell, and S. C. Mukhopadhyay, "Wireless sensor systems for space and extreme environments: A review," *IEEE Sensors J.*, vol. 14, no. 11, pp. 3955–3970, Nov. 2014.
- [2] P. Suriyachai, U. Roedig, and A. Scott, "A survey of MAC protocols for mission-critical applications in wireless sensor networks," *IEEE Commun. Surveys Tuts.*, vol. 14, no. 2, pp. 240–264, 2nd Quart., 2012.
- [3] C. Jendoubi and A. Asad, "A survey of artificial intelligence applications in nuclear power plants," *IoT*, vol. 5, no. 4, pp. 666–691, Oct. 2024.
- [4] A. Ayodeji, M. Mohamed, L. Li, A. D. Buono, I. Pierce, and H. Ahmed, "Cyber security in the nuclear industry: A closer look at digital control systems, networks and human factors," *Prog. Nucl. Energy*, vol. 161, Jul. 2023, Art. no. 104738.
- [5] A. B. de Neira, B. Kantarci, and M. Nogueira, "Distributed denial of service attack prediction: Challenges, open issues and opportunities," *Comput. Netw.*, vol. 222, Feb. 2023, Art. no. 109553.
- [6] T. Record. (2023). *Ukraine's State-Owned Nuclear Power Operator Said Russian Hackers Attacked Website*. [Online]. Available: <https://therecord.media/ukraines-state-owned-nuclear-power-operator-said-russian-hackers-attacked-website>
- [7] C. Hub. (2023). *Sellafield Nuclear Site Attacked by Cyber Groups Linked to Russia and China*. [Online]. Available: <https://www.cshub.com/attacks/news/sellafield-nuclear-site-attacked-by-cyber-groups-linked-to-russia-and-china>
- [8] T. C. Express. (2023). *Garnesia Team Claims Npcil Cyberattack*. [Online]. Available: <https://theyberexpress.com/npcil-cyberattack-by-garnesia-team/>
- [9] A. Greenberg. (2019). *The Wired Guide to Cyberwar*. [Online]. Available: <https://www.wired.com/story/cyberwar-guide/>
- [10] M. P. Arthur, G. Ramachandran, K. Sood, P. Kaarthik, S. Sridhar, and M. Chowdhury, "Empirical study of hierarchical intrusion detection systems for unknown attacks," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 6, pp. 5564–5581, Dec. 2025.
- [11] M. L. Ali, K. Thakur, S. Schmeelk, J. DeBello, and D. Dragos, "Deep learning vs. machine learning for intrusion detection in computer networks: A comparative study," *Appl. Sci.*, vol. 15, no. 4, p. 1903, Feb. 2025.
- [12] M. Mendoza and P. V. Tsvetkov, "An intelligent fault detection and diagnosis monitoring system for reactor operational resilience: Unknown fault detection," *Prog. Nucl. Energy*, vol. 171, Jun. 2024, Art. no. 105167.
- [13] A. Salehpour and I. Al-Anbagi, "Digital substations: Cyberattack detection system for small modular reactor-based power plants," *IEEE Electr. Mag.*, vol. 12, no. 4, pp. 57–67, Dec. 2024.
- [14] G. Wu, Y. Wang, and Z. Wu, "Physics-informed machine learning in cyber-attack detection and resilient control of chemical processes," *Chem. Eng. Res. Des.*, vol. 204, pp. 544–555, Apr. 2024.
- [15] A. Zainudin, L. A. C. Ahakonye, R. Akter, D.-S. Kim, and J.-M. Lee, "An efficient hybrid-DNN for DDoS detection and classification in software-defined IIoT networks," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8491–8504, May 2023.
- [16] S. Mohammadi, A. Balador, S. Sinaei, and F. Flammini, "Balancing privacy and performance in federated learning: A systematic literature review on methods and metrics," *J. Parallel Distrib. Comput.*, vol. 192, Oct. 2024, Art. no. 104918.
- [17] S. Wang, S. Hosseinalipour, M. Gorlatova, C. G. Brinton, and M. Chiang, "UAV-assisted online machine learning over multi-tiered networks: A hierarchical nested personalized federated learning approach," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 2, pp. 1847–1865, Jun. 2023.
- [18] A. Zainudin, R. Akter, D.-S. Kim, and J.-M. Lee, "Federated learning inspired low-complexity intrusion detection and classification technique for SDN-based industrial CPS," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 3, pp. 2442–2459, Sep. 2023.
- [19] T. Wang, X. Huang, Y. Wu, L. Qian, B. Lin, and Z. Su, "UAV swarm-assisted two-tier hierarchical federated learning," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 1, pp. 943–956, Jan. 2024.
- [20] S. Iqbal et al., "Hierarchical continual learning for domain-knowledge retention in healthcare federated learning," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 5025–5035, May 2025.
- [21] Z. Ullah et al., "Hybrid CNN-LSTM model for DDoS attack detection in Internet of Things-based healthcare Industry 5.0," *IEEE Internet Things J.*, vol. 12, no. 22, pp. 46075–46082, Nov. 2025.
- [22] J. Bhayo, R. Jafaq, A. Ahmed, S. Hameed, and S. A. Shah, "A time-efficient approach toward DDoS attack detection in IoT network using SDN," *IEEE Internet Things J.*, vol. 9, no. 5, pp. 3612–3630, Mar. 2022.
- [23] G. Wu, H. Zhang, W. Wu, Y. Wang, and Z. Wu, "Physics-informed graph convolutional recurrent network for cyber-attack detection in chemical process networks," *Ind. Eng. Chem. Res.*, vol. 64, no. 6, pp. 3370–3382, Feb. 2025.
- [24] Kamaldeep, M. Malik, and M. Dutta, "Feature engineering and machine learning framework for DDoS attack detection in the standardized Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8658–8669, May 2023.
- [25] S. D. Vyas, S. Kumar Padisala, and S. Dey, "A physics-informed neural network approach towards cyber attack detection in vehicle platoons," in *Proc. Amer. Control Conf. (ACC)*, May 2023, pp. 4537–4542.
- [26] G. B. Gaggero, P. Girdinio, and M. Marchese, "Artificial intelligence and physics-based anomaly detection in the smart grid: A survey," *IEEE Access*, vol. 13, pp. 23597–23606, 2025.
- [27] L. Lei, Y. He, Z. Xing, Z. Li, and Y. Zhou, "Physics-informed LSTM-based time-series forecasting model for power transformers," *IEEE Trans. Ind. Informat.*, vol. 21, no. 7, pp. 5411–5419, Jul. 2025.
- [28] X. Wu, P. Li, Y. Gu, J. Tao, S. Shen, and S. Yu, "Improved Gale-Shapley-based hierarchical federated learning for IoT scenarios," *IEEE Internet Things J.*, vol. 12, no. 7, pp. 9195–9205, Apr. 2025.
- [29] H. Devaraj et al., "RuralAI in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning for crop health monitoring," *IEEE Sensors Lett.*, vol. 8, no. 5, pp. 1–4, May 2024.
- [30] A. Ayodeji, A. Di Buono, I. Pierce, and H. Ahmed, "Wavy-attention network for real-time cyber-attack detection in a small modular pressurized water reactor digital control system," *Nucl. Eng. Design*, vol. 424, Aug. 2024, Art. no. 113277.
- [31] R. Rockabrand, "How artificial intelligence will power emergency preparedness and response for small modular reactors," Ph.D. dissertation, Dept. Artif. Intell., Capitol Technol. Univ., Laurel, MD, USA, 2025.

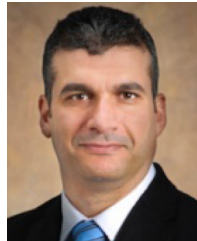
- [32] K. Nam, K. Kwon, and A. Kim, "HINT-sec: Hardware-in-the-loop nuclear power plant testbed for cyber security," *Prog. Nucl. Energy*, vol. 180, Feb. 2025, Art. no. 105600.
- [33] B. Zhang, S. Wang, S. Cheng, J. Sun, M. Peng, and C. Wang, "Transient trend prediction of safety parameters for small modular reactor considering equipment degradation," *Ann. Nucl. Energy*, vol. 181, Feb. 2023, Art. no. 109507.
- [34] International Atomic Energy Agency. (2024). *Small Modular Reactors: Advances in SMR Developments 2024*. Vienna, Non-Serial Publications. [Online]. Available: https://aris.iaea.org/Publications/SMR_catalogue_2024.pdf
- [35] World Nuclear Association. (2024). *Small Nuclear Power Reactors*. Accessed: Jun. 2, 2025. [Online]. Available: <https://world-nuclear.org/information-library/nuclear-fuel-cycle/nuclear-power-reactors/small-nuclear-power-reactors>
- [36] International Atomic Energy Agency, *Technology Roadmap for Small Modular Reactor Deployment*, document NR-T-1.18, IAEA Nuclear Energy Series, IAEA, Vienna, Austria, 2021. [Online]. Available: https://www-pub.iaea.org/MTCD/Publications/PDF/PUB1944_web.pdf
- [37] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, Jun. 2023.
- [38] U.S. Nuclear Regulatory Commission. (2025). *Digital Instrumentation and Controls Research*. Accessed: May 18, 2025. [Online]. Available: <https://www.nrc.gov/about-nrc/regulatory/research/digital.html>
- [39] International Atomic Energy Agency. (Dec. 2021). *Small Modular Reactors: Design Features, Safety Approaches and Research Programs*. International Atomic Energy Agency (IAEA). Accessed: May 18, 2025. [Online]. Available: https://www.iaea.org/sites/default/files/19/12/smr_rf_dsa_interim_report.pdf
- [40] NuScale Power. (2025). *Design Documents*. Accessed: May 18, 2025. [Online]. Available: <https://www.nuscalepower.com/technology/design-documents>
- [41] Standards Council of Canada. (2025). *Cybersecurity Standard CSA t100*. Accessed: May 18, 2025. [Online]. Available: <https://scc-ccn.ca/standardsdb/standards/2030549>
- [42] I. Al-Anbagi, M. Erol-Kantarci, and H. T. Mouftah, "Delay-aware medium access schemes for WSN-based partial discharge measurement," *IEEE Trans. Instrum. Meas.*, vol. 63, no. 12, pp. 3045–3057, Dec. 2014.
- [43] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.
- [44] A. Polanco and M. E. J. Newman, "Hierarchical core-periphery structure in networks," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 108, no. 2, Aug. 2023, Art. no. 024311.
- [45] Canadian Nuclear Safety Commission. (2025). *Dis-16-04: Small Modular Reactors-Regulatory Strategy, Approaches and Challenges*. Canadian Nuclear Safety Commission, Discussion Paper DIS-16-04. [Online]. Available: <https://www.cnsccn.gc.ca/eng/acts-and-regulations/consultation/comment/d-16-04/>
- [46] North American Electric Reliability Corporation (NERC). (Feb. 2022). *Provincial Summary-Saskatchewan. PDF Document*. [Online]. Available: <https://www.nerc.com/AboutNERC/keyplayers/ProvincialSummaries/Saskatchewan.pdf>
- [47] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020.
- [48] H. Liu, Y. Ma, C. Guo, X. Liu, and T. Wang, "MOHFL: Multi-level one-shot hierarchical federated learning with enhanced model aggregation over non-IID data," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 3, pp. 2853–2865, Jun. 2025.
- [49] I. Sharafaldin, A. H. Lashkari, S. Hakak, and A. A. Ghorbani, "Developing realistic distributed denial of service (DDoS) attack dataset and taxonomy," in *Proc. Int. Carnahan Conf. Secur. Technol. (ICCTST)*, Oct. 2019, pp. 1–8.



Md Facklasur Rahaman received the B.Sc. degree in electrical and electronic engineering from Rajshahi University of Engineering and Technology, Bangladesh, in 2016, and the Master of Engineering degree in IT convergence engineering from the Kumoh National Institute of Technology, Gumi, South Korea, in 2025. He is currently pursuing the Ph.D. degree in electronic systems engineering with the University of Regina, Canada. His research interests include federated learning, blockchain, metaverse, the Internet of Things, and large language models.



Makhduma F. Saiyed (Member, IEEE) received the Ph.D. degree in electronic systems engineering from the University of Regina, Regina, SK, Canada, in 2025. She was a Post-Doctoral Fellow with the University of Regina from March 2025 to June 2025. Currently, she is an Assistant Professor with the Department of Computer Science, Trent University, Oshawa, ON, Canada. She is also a registered Engineer-in-Training (EIT) with the Association of Professional Engineers and Geoscientists of Saskatchewan (APEGS), Regina. She has over ten years of teaching experience and has published numerous research articles. Her current research interests include network security, the Internet of Things (IoT), machine learning, deep neural networks, game theory, explainable AI, and large language models.



Irfan Al-Anbagi (Senior Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Ottawa in October 2013. From 2013 to 2015, he was a Post-Doctoral Fellow and the Product Development Manager of the Sec-Charge Project with the University of Ottawa. He is currently a Professor with the Faculty of Engineering and Applied Science, University of Regina, and an Adjunct Professor with the Department of Electrical and Computer Engineering, College of Engineering, University of Saskatchewan. He is also registered as a Professional Engineer with the Association of Professional Engineers and Geoscientists of Saskatchewan (APEGS) and a Professional Engineers Ontario (PEO). His research interests include security and reliability in cyber-physical systems, the Internet of Things (IoT) systems, and edge computing.



Ramakrishna Gokaraju (Senior Member, IEEE) received the B.E. degree (Hons.) in electrical and electronics engineering from the National Institute of Technology, Tiruchirappalli, India, in 1992, and the M.Sc. and Ph.D. degrees in electrical and computer engineering from the University of Calgary, Canada, in 1996 and 2000, respectively. He was a Research Scientist with Alberta Research Council and a Staff Software Engineer with the IBM Toronto Laboratory. Since 2003, he has been with the University of Saskatchewan, where he is currently a Professor and a Graduate Chair with the Department of Electrical and Computer Engineering. His research interests include power system protection and control and nuclear modeling of small modular reactors (SMRs). His research was funded by the NSERC Discovery Grant and the NSERC-Canada Nuclear Safety Commission (CNSC) Small Modular Reactor Research Grant.